Student Seminar - IDP Presentation 6

Can Ersoy - 03.11.2026

1) Why do some papers have no extractions even though they were classified as relevant?

- Some keywords trigger relevancy
  - Example: “self-assembly” and “tetrapeptide” in abstract/title but paper is about AI model and doesn’t mention any actual experiment data (lower confidence but still relevance classification)

- Currently, tables and images are left out of what is sent to the LLM

- 9/38 papers actually got at least one entry in a second extraction run
  (1 of these papers had 14 entries with length=1)

2) How can there be extractions with length=1

- Extractions leave out termini from sequences:
  - Fmoc-Lys(Fmoc)-OH -> **sequence: K** (Lysine - Lys)
  - Fmoc-Glu -> **sequence: E** (Glutamic Acid - Glu)

- Some papers use numbers to refer to multiple peptides to make it easier for the reader (e.g peptide 1 -> sequence:1)
  - Current field description: **Peptide sequence in letters. (e.g. 'FF')**
  - Maybe something like: Peptide sequence in valid IUPAC single character coding

3) What distinguishes papers with high extraction counts from the others?

- Some papers directly give the IUPAC encodings, making it easier for the LLM to extract

- Some papers compare different sequences, meaning the information is close to each other allowing LLM to extract easily

- Some papers give more information in tables/images and not in text (same as Q1)

New Queries (db=pmc instead of db=pubmed)

- Basically “peptide self-assembly” + year information due to api return limit of 10k (in total ~55k results) + pmc open access subset

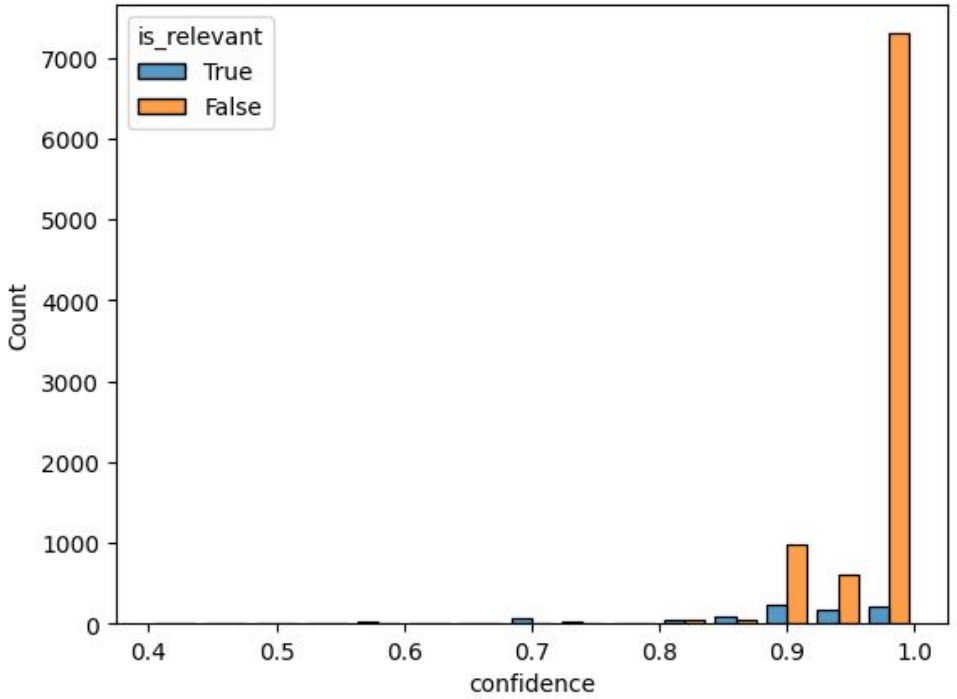
”("peptid"[All Fields] OR "peptidal"[All Fields] OR "peptide s"[All Fields] OR "peptides"[Supplementary Concept] OR "peptides"[All Fields] OR "peptide"[All Fields] OR "peptides"[MeSH Terms] OR "peptidic"[All Fields]) AND "self-assembly"[All Fields] AND (open_access[Filter]) AND ({}:{}[pdat]) AND (pmc_public[Filter])”

Relevance Check (first ~10k)

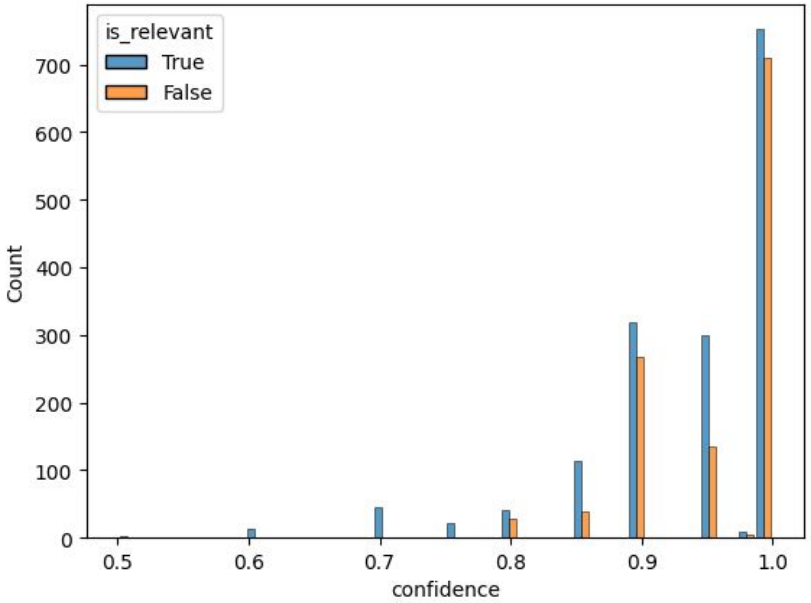
Model: gemma4:26b (MoE)
Time: ~13h

is_relevant (9850):

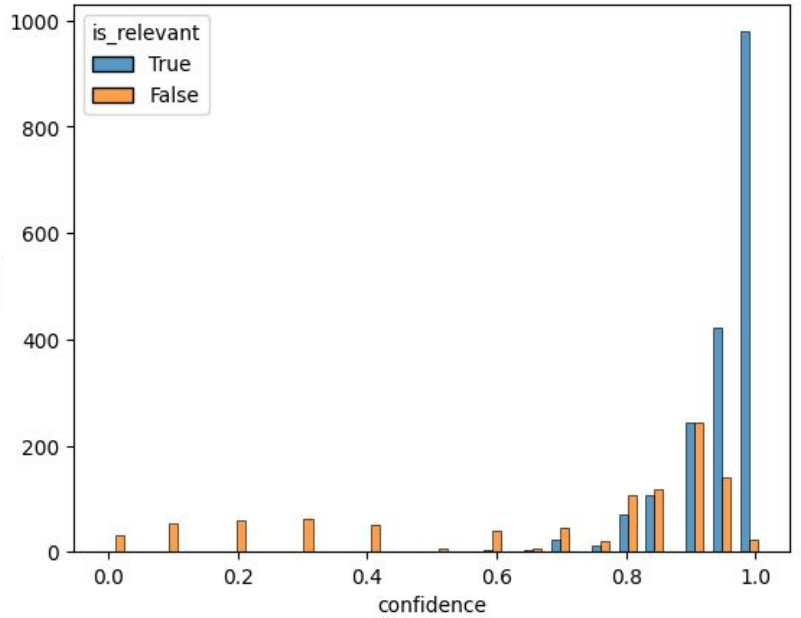
- **True: 881 papers**
- **False: 8969 papers**



Comparison (Gemini 3.5 Flash Lite and Gemma 4 26B)



Gemma 4 26b



Gemini 3.5 Flash Lite

Notes on LLMs

- Tested models around ~120b size, increasing parameter count improves extraction
- Dense models same size seem to perform better (for extraction task) than MoE, but are slower
- Speed is mostly determined by whether the model fits onto a single GPU or not (also model size)

Next Steps

- Test assumptions (e.g enforce IUPAC to lower sequence extraction errors…)
- Add tables (TableNode) to passed texts and compare extraction performance for table heavy papers
- Add improved logging for better error handling